

Assignment: Windows Server

Module 12: Installation, Storage, and Compute with Windows Server

1. What two options are provided in the type of installation window during Windows Server 2016 installation?

Ans: In Windows Server 2016 installation, two options are available:

- Upgrade: Install Windows and keep files, settings and applications
- Custom: Install Windows only (Advanced)

2. Write the step How to configure server step by step?

Ans: After installing Windows Server 2016:

- Set Administrator password.
- Rename server.
- Configure IP Address:
 - Open Network Settings
 - Assign Static IP
 - Set Subnet Mask
 - Configure Default Gateway
 - Set DNS Server
- Set Date and Time.
- Install required roles and features:
 - AD DS
 - DNS
 - DHCP
 - File Services
- Join server to domain (if needed).
- Configure firewall settings.
- Install updates.
- Create users and groups.
- Configure backup.

3. What are the Pre installation tasks?

Ans: Before installing server:

- Check hardware requirements
- Verify compatibility
- Backup important data
- Prepare installation media (USB/DVD)
- Decide server edition (Standard/Datacenter)
- Plan partitioning
- Assign IP addressing plan
- Collect product key
- Check BIOS/UEFI settings

4. What are the Post installation tasks?

Ans: Tasks after installation:

- Activate Windows
- Install drivers
- Set static IP
- Install updates
- Configure roles/features
- Promote as Domain Controller (if needed)
- Configure users and security
- Enable backups
- Monitor logs

5. What is the standard upgrade path for Windows Server?

Ans: Typical upgrade path:

- Windows Server 2008 → Windows Server 2012 → Windows Server 2016
- Windows Server 2012 R2 → Windows Server 2016

6. What is the Physical structure of AD?

Ans: Physical components:

- Domain Controllers (DCs)

- Sites
- Subnets

Purpose: Controls replication and network traffic.

7. What is the Logical components of Active Directory?

Ans: Logical structure includes:

➤ **Domain**

Administrative boundary.

Example:

Company.com

➤ **Tree**

Multiple related domains.

➤ **Forest**

Collection of domains.

➤ **OU**

Department grouping.

HR

IT

Sales

8. What is the Full form Of LDAP?

Ans: **LDAP** = Lightweight Directory Access Protocol.

9. What is the location of the AD database?

Ans: **AD database file:**

➤ C:\Windows\NTDS\NTDS.dit

- Stores users, groups, policies.

10. *What is child DC?*

Ans: **A Child Domain Controller manages a child domain under parent domain.**

- Example:
Parent: company.com
Child: sales.company.com

11. Explain the term forest in AD

Ans: A Forest is the highest level in Active Directory.
It contains:

- One or more domains
- Shared schema
- Shared global catalog
- Trust relationships

➤ Example:

company.com
sales.company.com
hr.company.com
All together = Forest.

12. What is Active Directory? Check all that apply.

- An open-source directory server
 - A Windows-only implementation of a directory server
 - Microsoft's implementation of a directory server
 - An LDAP-compatible directory server
- **Ans:** Active Directory is used to **store, organize, and manage** information about resources on a network

- ✓ A Windows-only implementation of a directory server
- ✓ Microsoft's implementation of a directory server
- ✓ An LDAP-compatible directory server

13. When you create an Active Directory domain, what's the name of the default user account?

- Superuser
- Root
- Username
- Administrator

Ans: Administrator

14. AD domain provides which of the following advantages? Check all that apply

- Centralized authentication
- More detailed logging
- Centralized management with GPOs
- Better performance

Ans:

- Centralized authentication
- More detailed logging
- Centralized management with GPOs

15. What are the minimum hardware requirements for installing Windows Server 2016?

Ans: For Windows Server 2016 :

- CPU: 1.4 GHz 64-bit
- RAM: 512 MB (2 GB recommended for GUI)
- Disk: 32 GB
- Network Adapter: Gigabit Ethernet

DVD/USB boot media

16. Explain the different editions of Windows Server 2016 and their features.

Ans: a) **Standard Edition**

- Basic virtualization
- 2 virtual machines
- File server
- AD, DNS, DHCP

Used for small-medium businesses.

b) Datacenter Edition

- Unlimited VMs
- Shielded VMs
- Storage Spaces Direct
- Advanced networking

Used in data centers/cloud.

c) Essentials

- Small organizations
- Limited users
- Simpler management

17. Walk through the steps of installing Windows Server 2016 using GUI mode.

Ans:

- Boot from DVD/USB
- Select Language
- Click Install Now
- Enter product key
- Select **Server with Desktop Experience**
- Accept license
- Choose Custom Install
- Select disk
- Click Next
- Installation starts
- Set Administrator password
- Login and configure server

18. Describe the steps for installing Windows Server 2016 in Server Core mode.

Ans: Windows Server 2016 Standard (Server Core Installation)

No GUI installed.

Configuration done using:

- PowerShell
- Command line
- sconfig tool

Why use Core?

- Less resources
- Better security

Fewer updates

19. How do you configure network settings during Windows Server 2016 installation?

Ans: Steps:

- Open Network Settings
- Change Adapter Settings
- Open IPv4 Properties
- Assign:

Static IP
Subnet Mask
Gateway
DNS

➤ Example:

IP: 192.168.1.10
Mask: 255.255.255.0
Gateway: 192.168.1.1
DNS: 192.168.1.10

20. Explain the process of promoting a Windows Server to a domain controller.

Ans:

- Open Server Manager
- Add Roles and Features
- Install AD DS role
- Click Promote this server to domain controller
- Choose: New Forest OR

Add Domain Controller

- Set domain name
- Set DSRM password
- Install DNS (optional)
- Click Install
- Server reboots

21. Discuss the steps involved in upgrading from a previous version of Windows Server to Windows Server 2016.

Ans:

- Backup system
- Check compatibility
- Mount installation media
- Run setup.exe
- Select Upgrade
- Keep files/apps
- Start upgrade
- Reboot and verify services

22. What is Active Directory Domain Services (AD DS), and what are its key components?

Ans: AD DS (Active Directory Domain Services) stores and manages:

- Users
- Computers
- Groups
- Policies
- Security

Key components:

- Domain Controllers
- Domains
- Forests
- OUs
- Global Catalog
- Schema

23. How do you create a new Active Directory user account in Windows Server ?

Ans:

- Open **Active Directory Users and Computers**

- Select OU
- Right-click → New → User
- Enter:
 - Name
 - Username
 - Password
- Finish

24. Explain the process of creating and managing Group Policy Objects (GPOs) in Windows Server 2016 or 2019.

Ans: Using Group Policy Management:

- Open Group Policy Management
- Create New GPO
- Edit settings
 - Examples:
 - Password policy
 - USB block
 - Desktop restrictions
- Link GPO to OU/domain
- Run:
 - gpupdate /force
- **Why use GPO?**
Centralized management.

25. What are Organizational Units (OUs) in Active Directory, and how do you use them?

Ans: OU = Organizational Unit

Used to organize objects.

Example:

Company

├— HR

├— IT

└─ Sales

Uses:

- Organize users
- Apply GPOs
- Delegate admin rights

26. Describe the process of delegating administrative privileges in Active Directory.

Ans: OU = Organizational Unit

Used to organize objects.

Example:

Company

└─ HR

└─ IT

└─ Sales

Uses:

- Organize users
- Apply GPOs
- Delegate admin rights